

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 7

Urinary System

Chapter Goals (Slide 1 of 2)

- Name essential organs of the urinary system and describe their locations and functions.
- Identify common pathological conditions.
- Recognize how urinalysis is used and interpreted as a diagnostic test.
- Define urinary-system-related combining forms, prefixes, and suffixes.

Chapter Goals (Slide 2 of 2)

- List and explain clinical procedures, laboratory tests, and abbreviations that pertain to the urinary system.
- Understand medical terms in their proper contexts, such as medical reports and records.

Chapter 7

Lesson 7.1

Introduction (Slide 1 of 2)

- **Nitrogenous wastes**
 - **Urea**
 - **Creatinine**
 - **Uric acid**

Introduction (Slide 2 of 2)

- **Functions of the kidney**

- Filter nitrogenous wastes to form urine; about 200 quarts of blood are filtered every day to form 2 quarts of urine
- Maintain proper balance of water, electrolytes (sodium, potassium), and acids
- Release hormones:
 - **Renin:** enzyme important in adjusting blood pressure
 - **Erythropoietin (EPO):** hormone that stimulates red blood cell production in the bone marrow
 - **Calciferol:** active form of vitamin D necessary for the absorption of calcium from the intestine
- Degrade and eliminate hormones from the bloodstream

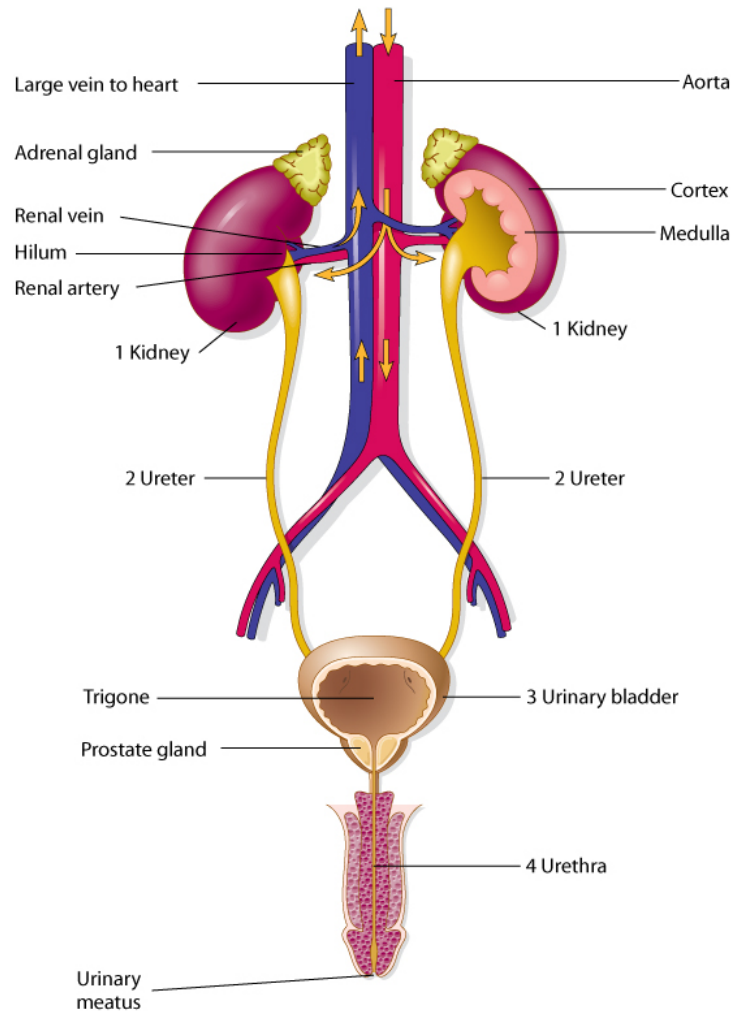
Anatomy of the Major Organs (Slide 1 of 4)

Organs of the male urinary system:

- **Kidney**
- **Ureter**
- **Urinary bladder**
- **Urethra**

Anatomy of the Major Organs (slide 2 of 4)

Organs of the male urinary system



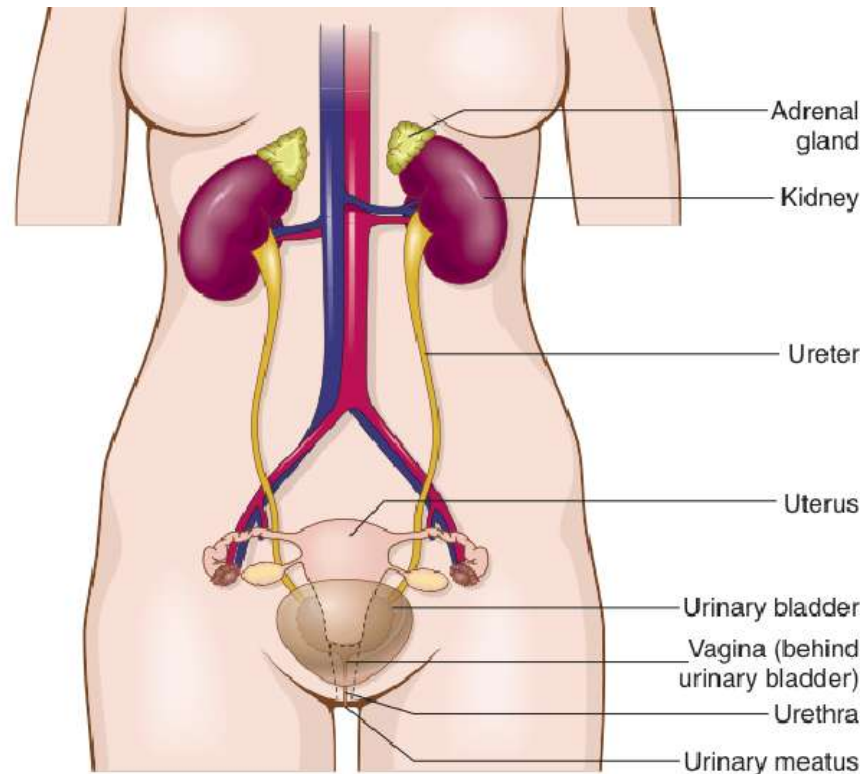
Anatomy of the Major Organs (slide 3 of 4)

Organs of the female urinary system:

- **Kidney**
- **Ureter**
- **Urinary bladder**
- **Urethra**

Anatomy of the Major Organs (slide 4 of 4)

Female urinary system



QUICK QUIZ (Slide 1 of 8)

1. What is voiding?

- A. Nitrogenous waste
- B. Urination
- C. Outer section of the kidney
- D. Triangular area in the bladder

How Kidneys Produce Urine

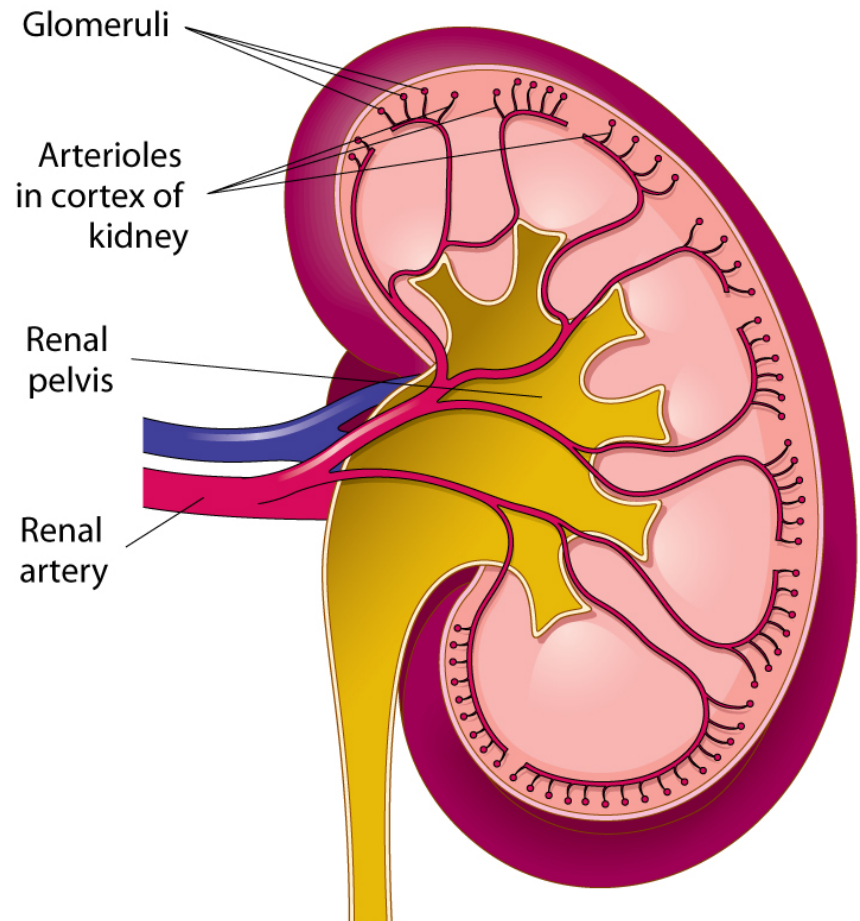
(Slide 1 of 12)

- Blood enters the kidneys through the right and left **renal arteries**
- **Arterioles** carry blood to the **capillaries**
- **Glomeruli** filter the blood

How Kidneys Produce Urine

(Slide 2 of 12)

- Blood enters the kidneys through the right and left **renal arteries**
- **Arterioles** carry blood to the **capillaries**
- **Glomeruli** filter the blood



How Kidneys Produce Urine

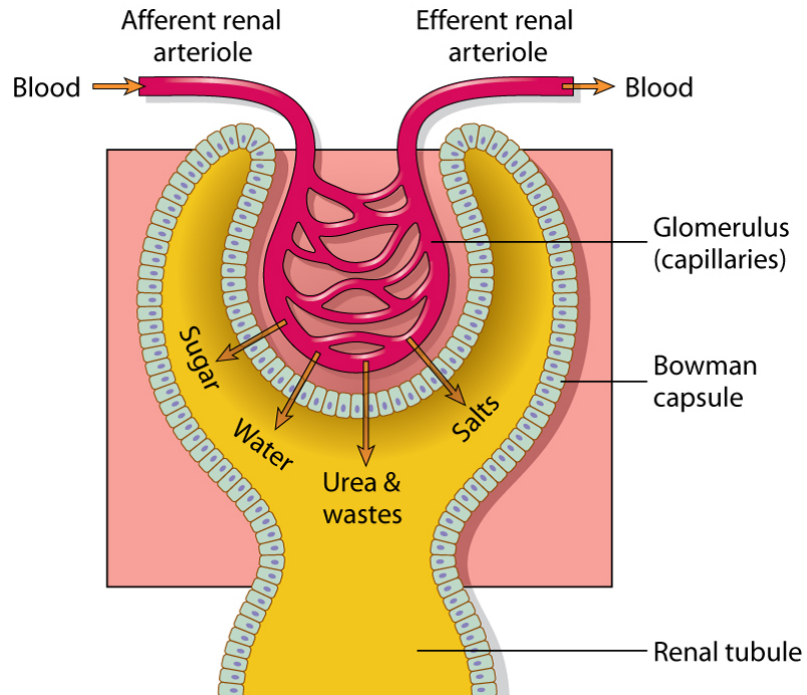
(Slide 3 of 12)

- **Glomerulus and glomerular/Bowman capsule**
- Blood passes through the glomeruli
- Glomerular (Bowman) capsule surrounds each glomerulus
- Renal tubule is attached to each Bowman capsule

How Kidneys Produce Urine

(Slide 4 of 12)

- **Glomerulus and glomerular/Bowman capsule**



How Kidneys Produce Urine

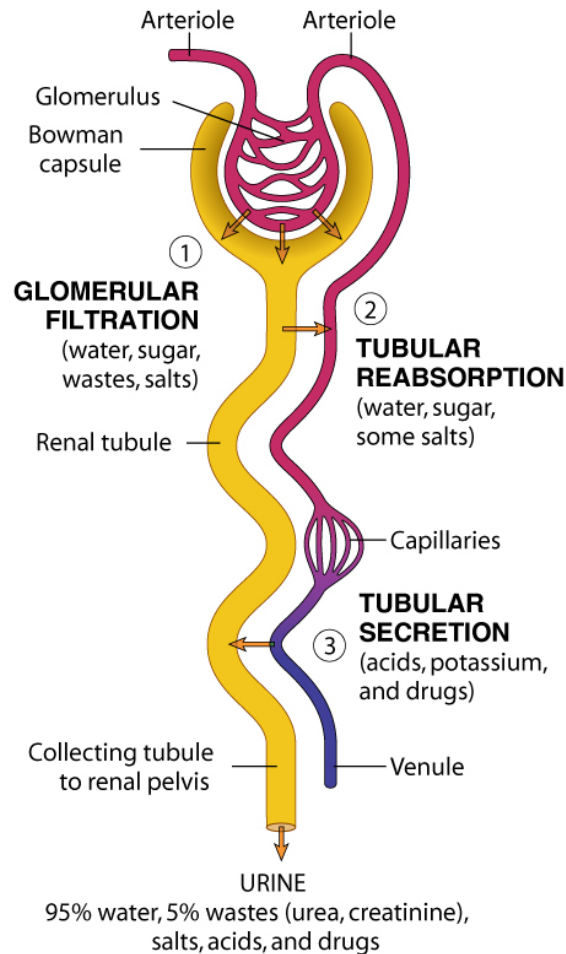
(Slide 5 of 12)

Three steps in the formation of urine

- Glomerular filtration (water, sugar, wastes, salts)
- Tubular reabsorption (water, sugar, sodium)
- Tubular secretion (acids, potassium, drugs)

How Kidneys Produce Urine

(Slide 6 of 12)



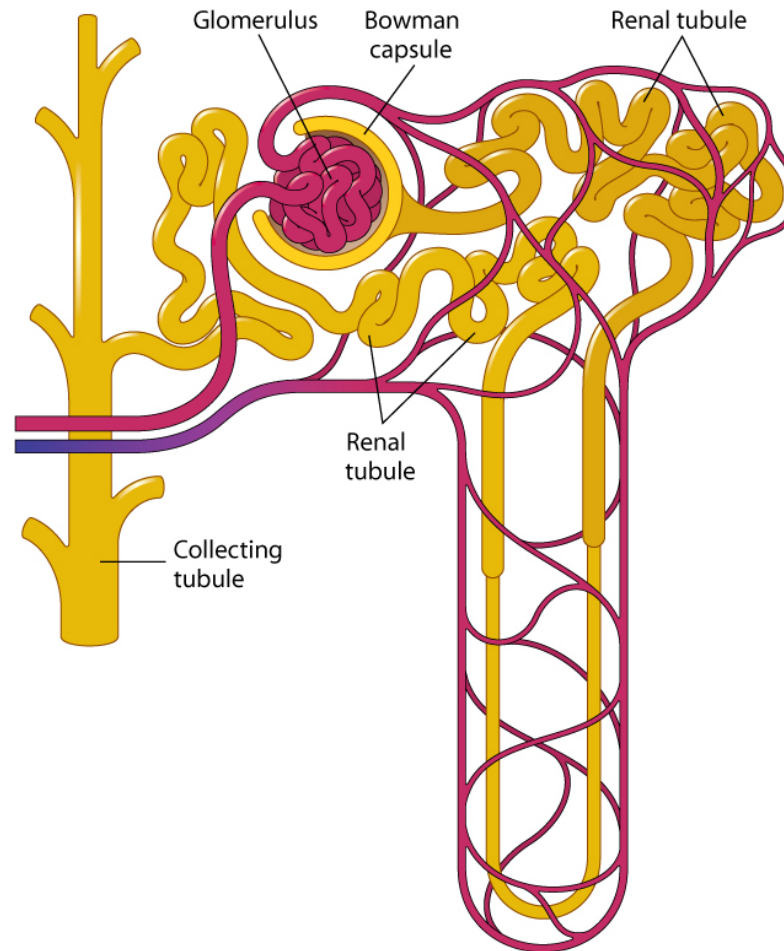
How Kidneys Produce Urine

(Slide 7 of 12)

- The glomerulus and a renal tubule combine to form a unit called a **nephron**.

How Kidneys Produce Urine

(Slide 8 of 12)



QUICK QUIZ (Slide 2 of 8)

2. Approximately how many nephrons are in a kidney?

A. 100,000,000

B. 10,000,000

C. 1,000,000

D. 100,000

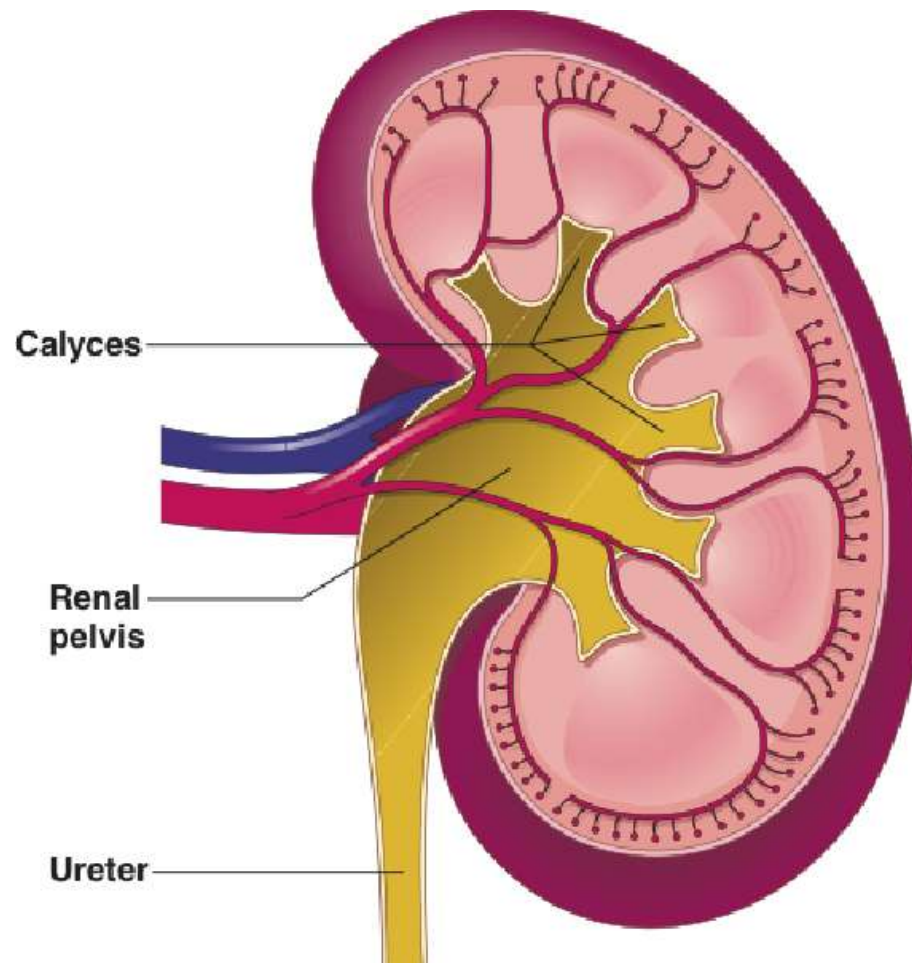
How Kidneys Produce Urine

(Slide 9 of 12)

- All collecting tubules lead to the **renal pelvis**.
- **Calyces or calices** are small, cuplike regions of the renal pelvis; the term comes from the Greek, *kalux*, meaning a cup or case surrounding a flower bud.

How Kidneys Produce Urine

(Slide 10 of 12)



How Kidneys Produce Urine

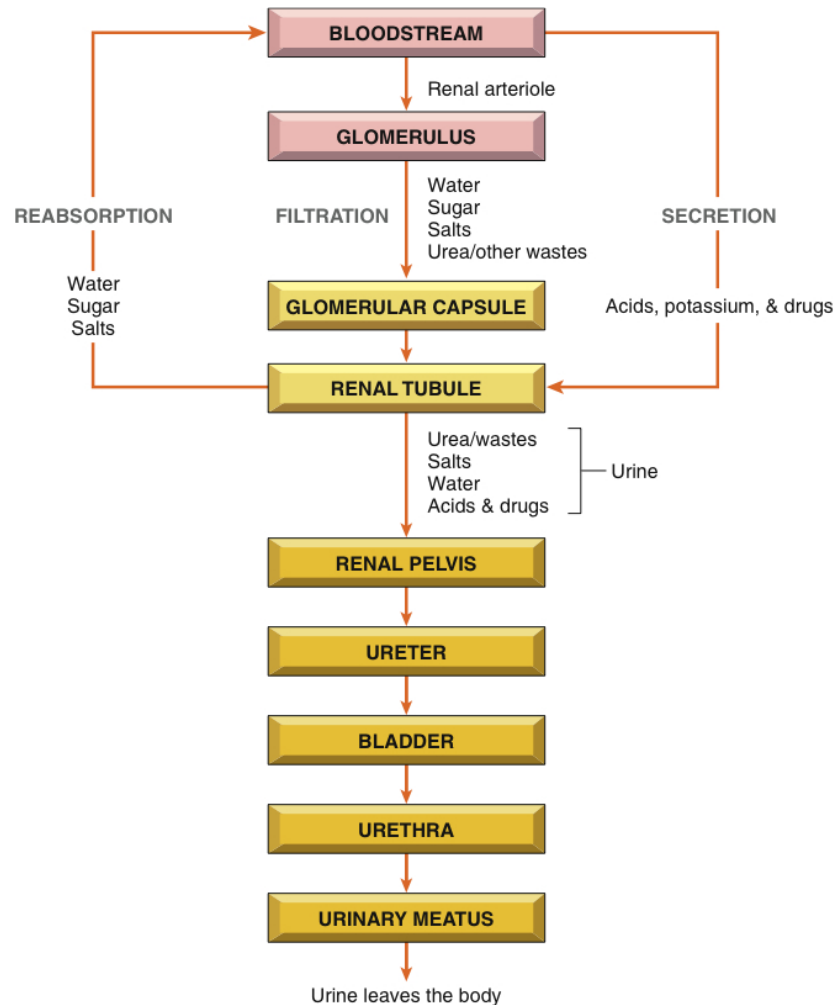
(Slide 11 of 12)

Functions of the Kidneys

- Remove nitrogenous wastes
- Balance water and electrolytes
- Release hormones
- Degrade and eliminate hormones

How Kidneys Produce Urine

(Slide 12 of 12)



Vocabulary (Slide 1 of 14)

Term
arteriole
calyx or calix
catheter
cortex
creatinine
electrolyte

Vocabulary (Slide 2 of 14)

Term	Meaning/Definition
arteriole	Small artery
calyx or calix	Cuplike collecting region of the renal pelvis; the term comes from the Greek, <i>kalux</i> , meaning a cup or case surrounding a flower bud
catheter	Tube for injecting or removing fluids
cortex	Outer region of an organ
creatinine	Nitrogenous waste excreted in urine
electrolyte	Chemical element that carries an electrical charge when dissolved in water.

Vocabulary (Slide 3 of 14)

Term
erythropoietin (EPO)
filtration
glomerular capsule
glomerulus
hilum

Vocabulary (Slide 4 of 14)

Term	Meaning/Definition
erythropoietin (EPO)	Hormone secreted by the kidney to stimulate production of red blood cells by bone marrow; -poietin means a substance that forms
filtration	Process whereby some substances pass through a filter
glomerular capsule	Enclosing structure surrounding each glomerulus; also called Bowman capsule
glomerulus	Tiny ball of capillaries in the kidney
hilum	Depression in the part of an organ where blood vessels and nerves enter and leave; comes from the Latin meaning <i>a small thing</i>

Vocabulary (Slide 5 of 14)

Term
kidney
meatus
medulla
nephron

Vocabulary (Slide 6 of 14)

Term	Meaning/Definition
kidney	One of two bean-shaped organs that filter nitrogenous waste from the bloodstream to form urine
meatus	Opening or canal
medulla	Inner region of an organ; the term comes from the Latin, <i>medulla</i> , meaning marrow
nephron	The functional unit of the kidney where filtration, reabsorption, and secretion take place

Vocabulary (Slide 7 of 14)

Term
nitrogenous waste
potassium (K ⁺)
reabsorption
renal artery

Vocabulary (Slide 8 of 14)

Term	Meaning/Definition
nitrogenous waste	Substance containing nitrogen and excreted in urine; examples are urea, uric acid, and creatinine
potassium (K ⁺)	An electrolyte regulated by the kidney
reabsorption	Renal tubules return materials necessary to the body back into the bloodstream
renal artery	Blood vessel that carries blood to the kidney

Vocabulary (Slide 9 of 14)

Term
renal pelvis
renal tubules
renal vein

Vocabulary (Slide 10 of 14)

Term	Meaning/Definition
renal pelvis	Central collection region in the kidney
renal tubules	Microscopic tubes in the kidney where urine is formed after filtration
renal vein	Blood vessel that carries blood away from the kidney

Vocabulary (Slide 11 of 14)

Term
renin
sodium (Na ⁺)
trigone
urea

Vocabulary (Slide 12 of 14)

Term	Meaning/Definition
renin	Enzyme secreted by the kidney that raises blood pressure
sodium (Na ⁺)	An electrolyte regulated in the blood and urine by the kidneys; a common form of sodium is sodium chloride (salt)
trigone	Triangular area in the urinary bladder
urea	Major nitrogenous waste excreted in urine

Vocabulary (Slide 13 of 14)

Term
ureter
urethra
uric acid
urinary bladder
urination (voiding)

Vocabulary (Slide 14 of 14)

Term	Meaning/Definition
ureter	One of two tubes leading from the kidneys to the urinary bladder
urethra	Tube leading from the urinary bladder to the outside of the body
uric acid	Nitrogenous waste product excreted in the urine
urinary bladder	Hollow, muscular sac that holds and stores urine
urination (voiding)	Process of expelling urine; also called micturition

Terminology:

Structures (Slide 1 of 6)

Combining Form

cali/o, calic/o

cyst/o

glomerul/o

meat/o

Terminology:

Structures (Slide 2 of 6)

Combining Form	Meaning
cali/o, calic/o	calyx (calix); cup-shaped
cyst/o	urinary bladder
glomerul/o	glomerulus
meat/o	meatus

Terminology:

Structures (Slide 3 of 6)

Combining Form

nephr/o

pyel/o

ren/o

trigon/o

Terminology:

Structures (Slide 4 of 6)

Combining Form	Meaning
nephr/o	kidney
pyel/o	renal pelvis
ren/o	kidney
trigon/o	trigone

Terminology:

Structures (Slide 5 of 6)

Combining Form

ureter/o

urethr/o

vesic/o

Terminology:

Structures (Slide 6 of 6)

Combining Form	Meaning
ureter/o	ureter
urethr/o	urethra
vesic/o	urinary bladder

Terminology:

Substances and Symptoms (Slide 1 of 6)

Combining Form

albumin/o

azot/o

bacteri/o

dips/o

Terminology:

Substances and Symptoms (Slide 2 of 6)

Combining Form	Meaning
albumin/o	albumin
azot/o	nitrogen
bacteri/o	bacteria
dips/o	thirst

Terminology:

Substances and Symptoms (Slide 3 of 6)

Combining Form

kal/o

ket/o, keton/o

lith/o

natr/o

noct/o

olig/o

Terminology:

Substances and Symptoms (Slide 4 of 6)

Combining Form	Meaning
kal/o	potassium
ket/o, keton/o	ketone bodies
lith/o	stone
natr/o	sodium
noct/o	night
olig/o	scanty

Terminology:

Substances and Symptoms (Slide 5 of 6)

Combining Form or Suffix

-poietin

py/o

-tripsy

ur/o

-uria

urin/o

Terminology:

Substances and Symptoms (Slide 6 of 6)

Combining Form or Suffix	Meaning
-poietin	substance that forms
py/o	pus
-tripsy	to crush
ur/o	urea
-uria	urination; urine condition
urin/o	urine

Urinalysis (Slide 1 of 2)

Tests Included in a Urinalysis

- 1. Color**
- 2. Appearance**
- 3. pH**
- 4. Protein**
- 5. Glucose**

Urinalysis (Slide 2 of 2)

Tests Included in a Urinalysis *(continued)*

- 6. Specific gravity
- 7. Ketone bodies
- 8. Sediment
- 9. Phenylketonuria
- 10. Bilirubin

QUICK QUIZ (Slide 3 of 8)

- 3. In a urinalysis, what does the test of specific gravity reflect?**
- A. The chemical nature of urine
 - B. The presence of albumin
 - C. Blood in the urine
 - D. The amount of wastes, minerals, and solids in the urine

Chapter 7

Lesson 7.2

Pathologic Conditions (Slide 1 of 5)

Kidney

- **Glomerulonephritis**
- **Interstitial nephritis**
- **Nephrolithiasis**
- **Nephrotic syndrome**

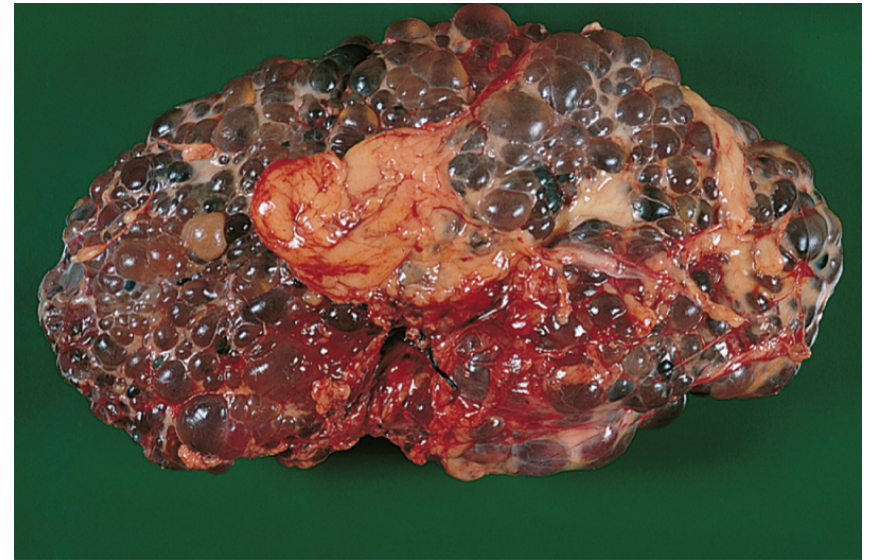
Pathologic Conditions

(Slide 2 of 5)

Kidney

- Polycystic kidney disease (PKD)

PKD—*The kidneys contain masses of cysts. Typically, polycystic kidneys weigh 20 times more than their usual weight.*



Pathologic Conditions (Slide 3 of 5)

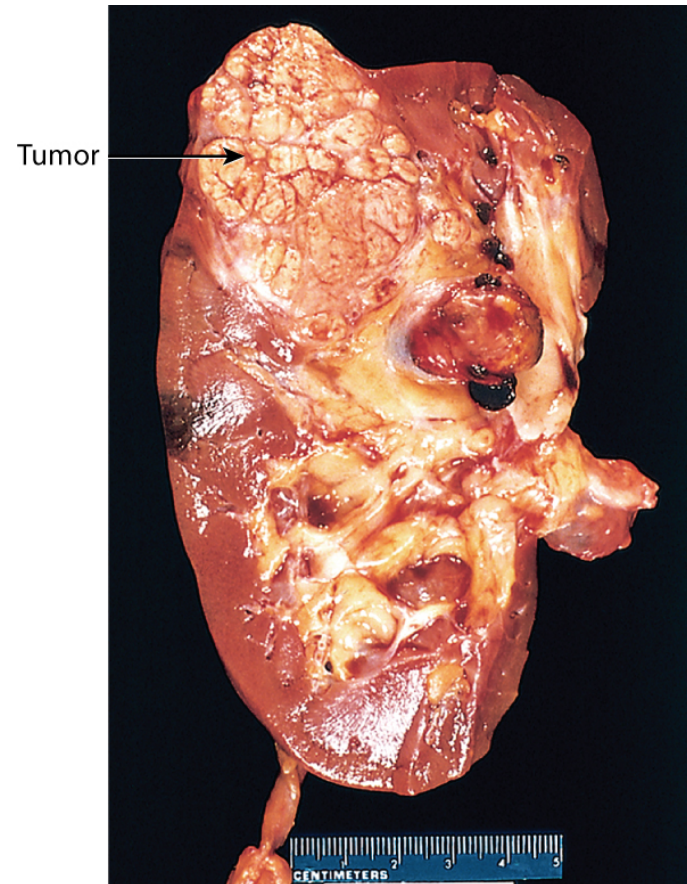
Kidney

- **Pyelonephritis**
- **Renal cell carcinoma**
- **Renal failure**
- **Renal hypertension**
- **Wilms tumor**

Pathologic Conditions (Slide 4 of 5)

Kidney

- Renal cell carcinoma



Pathologic Conditions (Slide 5 of 5)

Urinary bladder

- Bladder cancer

Associated conditions

- Diabetes insipidus (DI)
- Diabetes mellitus (DM)

QUICK QUIZ (Slide 4 of 8)

4. A term that means frequent (voluntary) urination at night is:

- A. anuria
- B. nocturia
- C. diuresis
- D. hematuria

Laboratory Tests and Clinical Procedures (Slide 1 of 10)

Laboratory tests

- Blood urea nitrogen (BUN)
- Creatinine clearance test

Laboratory Tests and Clinical Procedures (Slide 2 of 10)

Clinical procedures

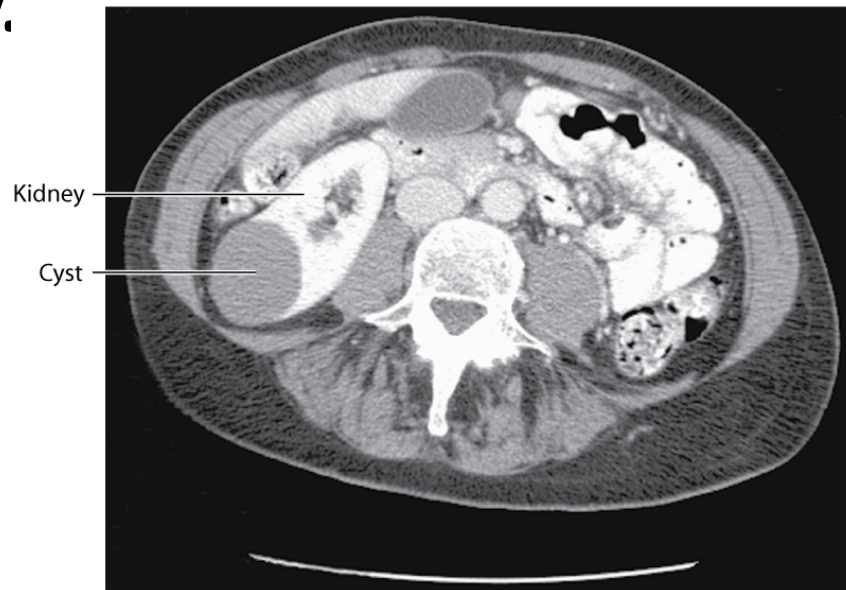
X-ray Studies

- **CT urography**
- **Kidneys, ureters, and bladder (KUB)**
- **Renal angiography**
- **Retrograde pyelogram (RP)**
- **Voiding cystourethrogram (VCUG)**

Laboratory Tests and Clinical Procedures (Slide 3 of 10)

- **CT-Computed tomography**

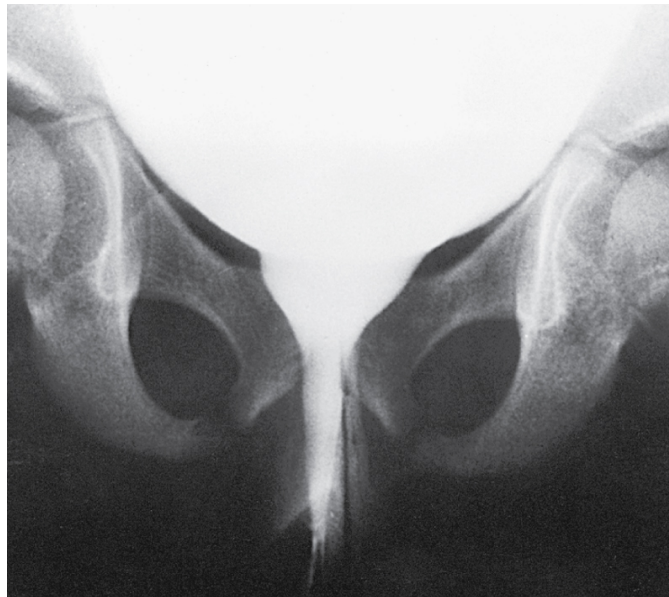
The scan with contrast shows a benign cyst on the kidney.



Laboratory Tests and Clinical Procedures (Slide 4 of 10)

Voiding cystourethrogram

- (VCUG) showing a normal female urethra. The bladder is filled with contrast material, followed by x-ray imaging.



Laboratory Tests and Clinical Procedures (Slide 5 of 10)

Clinical Procedures

Ultrasound examination

- **Ultrasonography:** imaging urinary tract structures using high frequency sound waves

Radioactive studies

- **Radioisotope scan:** image of kidney after injecting a radioisotope into the bloodstream

Laboratory Tests and Clinical Procedures (Slide 6 of 10)

Clinical Procedures

Magnetic imaging

- **Magnetic resonance imaging (MRI) urography:**
A changing magnetic field produces images of an organ and surrounding structures in three planes of the body

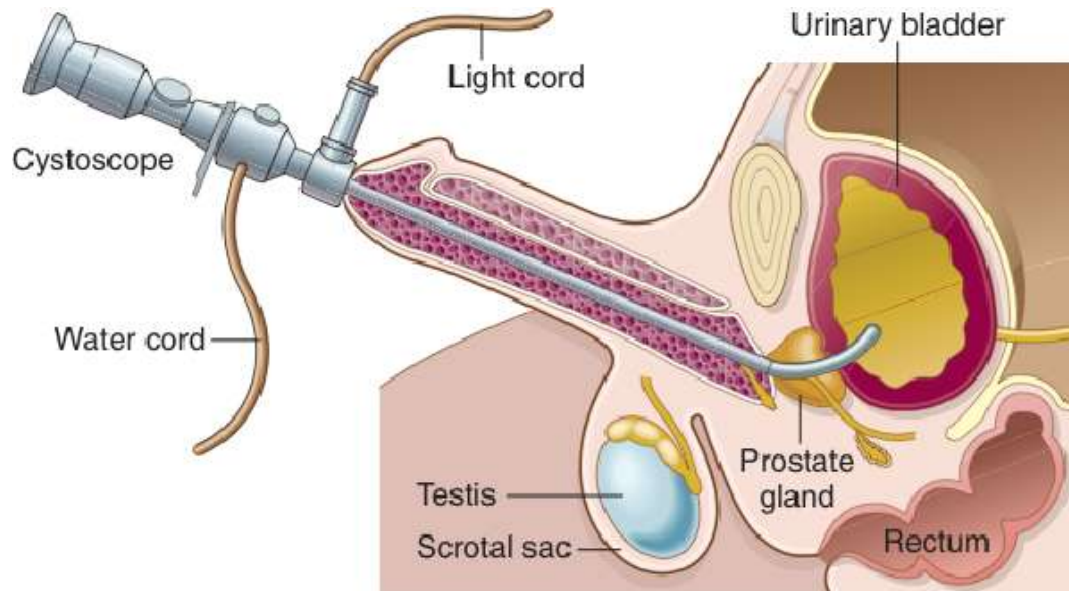
Laboratory Tests and Clinical Procedures (Slide 7 of 10)

Other Clinical Procedures

- Cystoscopy
- Dialysis
- Lithotripsy
- Renal angioplasty
- Renal biopsy
- Renal transplantation
- Urinary catheterization

Laboratory Tests and Clinical Procedures (Slide 8 of 10)

Cystoscopy—direct visual examination of urinary bladder with an endoscope



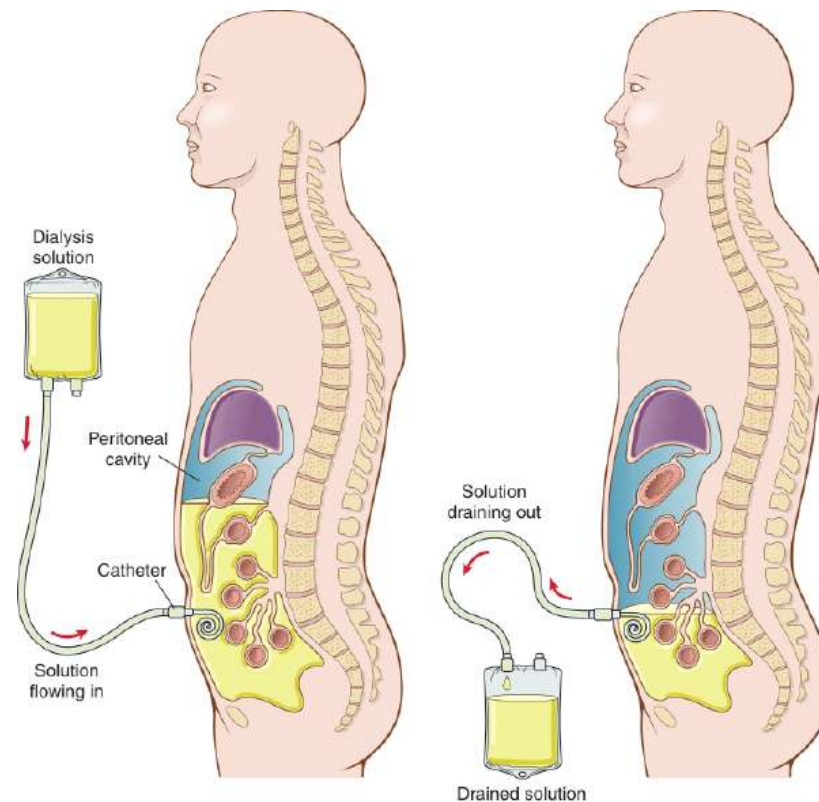
Laboratory Tests and Clinical Procedures (Slide 9 of 10)

Hemodialysis (HD)



Laboratory Tests and Clinical Procedures (Slide 10 of 10)

Continuous ambulatory peritoneal dialysis (CAPD)



Abbreviations (Slide 1 of 6)

Abbreviation

ADH

AKI

ARF

BILI

BUN

CAPD

Cath

CCPD

CKD

Abbreviations (Slide 2 of 6)

Abbreviation	Meaning
ADH	Antidiuretic hormone
AKI	Acute renal injury
ARF	Acute renal failure
BILI	Bilirubin
BUN	Blood urea nitrogen
CAPD	Continuous ambulatory peritoneal dialysis
Cath	Catheter; catheterization
CCPD	Continuous cycling peritoneal dialysis
CKD	Chronic kidney disease

Abbreviations (Slide 3 of 6)

Abbreviation

CL⁻

CrCl

CRF

C & S

Cysto

eGFR

ESRD

ESWL

HCO₃⁻

HD

IC

Abbreviations (Slide 4 of 6)

Abbreviation	Meaning
CL ⁻	Chloride, a kidney excretion
CrCl	Creatinine clearance
CRF	Chronic renal failure
C & S	Culture and sensitivity testing
Cysto	Cystoscopic examination
eGFR	Estimated glomerular filtration rate
ESRD	End-stage renal disease
ESWL	Extracorporeal shockwave lithrotripsy
HCO ₃ ⁻	Bicarbonate, an electrolyte conserved by the kidney
HD	Hemodialysis
IC	Interstitial cystitis, chronic inflammation of the bladder wall

Abbreviations (Slide 5 of 6)

Abbreviation

IVP

K⁺

KUB

Na⁺

PD

pH

PKD

sp gr

UA

UTI

VCUG

Abbreviations (Slide 6 of 6)

Abbreviation	Meaning
IVP	Intravenous pyelogram
K ⁺	Potassium, an electrolyte
KUB	Kidney ureter and bladder
Na ⁺	Sodium, an electrolyte
PD	Peritoneal dialysis
pH	Potential hydrogen; degree of acidity or alkalinity
PKD	Phenylketonuria
sp gr	Specific gravity
UA	Urinalysis
UTI	Urinary tract infection
VCUG	Voiding cystourethrogram

QUICK QUIZ (Slide 5 of 8)

5. The abbreviation RP is associated with which of the following?

- A. Fluid is injected into the peritoneal cavity and then drained out.
- B. Contrast is injected into the urinary bladder and ureters, and x-rays are taken of the urinary tract.
- C. Nitrogenous wastes are removed from the patient's blood.
- D. An electrolyte is secreted by renal tubules.

Review Sheet:

Combining Forms (Slide 1 of 10)

Combining Form
albumino/o
angi/o
azot/o
bacteri/o
cali/o, calic/o
cyst/o

Review Sheet:

Combining Forms (Slide 2 of 10)

Combining Form	Meaning
albumino/o	albumin (protein)
angi/o	vessel (blood)
azot/o	urea; nitrogen
bacteri/o	bacteria
cali/o, calic/o	calyx (calix); cup-shaped
cyst/o	urinary bladder

Review Sheet:

Combining Forms (Slide 3 of 10)

Combining Form
dips/o
glomerul/o
glycos/o
hydr/o
isch/o
kal/o
ket/o; keton/o

Review Sheet:

Combining Forms (Slide 4 of 10)

Combining Form	Meaning
dips/o	thirst
glomerul/o	glomerulus
glycos/o	sugar
hydr/o	water
isch/o	to hold back; back
kal/o	potassium
ket/o; keton/o	ketones; acetones

Review Sheet:

Combining Forms (Slide 5 of 10)

Combining Form
lith/o
meat/o
natr/o
necr/o
nephr/o
noct/o
ogli/o

Review Sheet:

Combining Forms (Slide 6 of 10)

Combining Form	Meaning
lith/o	stone
meat/o	meatus
natr/o	sodium
necr/o	death
nephr/o	kidney
noct/o	night
ogli/o	scanty

Review Sheet:

Combining Forms (Slide 7 of 10)

Combining Form
py/o
pyel/o
ren/o
trigon/o
ur/o

Review Sheet:

Combining Forms (Slide 8 of 10)

Combining Form	Meaning
py/o	pus
pyel/o	renal pelvis
ren/o	kidney
trigon/o	trigone
ur/o	urine; urinary tract

Review Sheet:

Combining Forms (Slide 9 of 10)

Combining Form
ureter/o
urethr/o
urin/o
vesic/o

Review Sheet:

Combining Forms (Slide 10 of 10)

Combining Form	Meaning
ureter/o	ureter
urethr/o	urethra
urin/o	urine
vesic/o	urinary bladder

Review Sheet:

Suffixes (Slide 1 of 8)

Suffix
-ectasis
-ectomy
-emia
-esis
-gram
-lithiasis

Review Sheet:

Suffixes (Slide 2 of 8)

Suffix	Meaning
-ectasis	stretching; dilation
-ectomy	removal; excision
-emia	blood condition
-esis	condition
-gram	record
-lithiasis	condition of stones

Review Sheet:

Suffixes (Slide 3 of 8)

Suffix
-lithotomy
-lysis
-megaly
-ole
-osis
-pathy

Review Sheet:

Suffixes (Slide 4 of 8)

Suffix	Meaning
-lithotomy	incision for removal of a stone
-lysis	breakdown
-megaly	enlargement
-ole	little; small
-osis	condition
-pathy	disease

Review Sheet:

Suffixes (Slide 5 of 8)

Suffix
-plasty
-poietin
-ptosis
-rrhea
-sclerosis
-stomy

Review Sheet:

Suffixes (Slide 6 of 8)

Suffix	Meaning
-plasty	surgical repair
-poietin	substance that forms
-ptosis	droop; sag
-rrhea	flow; discharge
-sclerosis	hardening
-stomy	new opening (to form a mouth)

Review Sheet:

Suffixes (Slide 7 of 8)

Suffix
-tomy
-tripsy
-uria

Review Sheet:

Suffixes (Slide 8 of 8)

Suffix	Meaning
-tomy	process of cutting
-tripsy	to crush
-uria	urination

Review Sheet:

Prefixes (Slide 1 of 4)

Prefix
a-, an-
anti-
dia-
dys-

Review Sheet:

Prefixes (Slide 2 of 4)

Prefix	Meaning
a-, an-	not; without
anti-	again
dia-	complete
dys-	bad; painful

Review Sheet:

Prefixes (Slide 3 of 4)

Prefix
en-
peri-
poly-
retro-

Review Sheet:

Prefixes (Slide 4 of 4)

Prefix	Meaning
en-	in; within
peri-	surrounding
poly-	many, much
retro-	behind; back

QUICK QUIZ (Slide 6 of 8)

6. The combining form dips/o means:

- A. stone
- B. scanty
- C. thirst
- D. water

QUICK QUIZ (Slide 7 of 8)

7. Which combining form means pus?

A. pyel/o

B. lith/o

C. ogli/o

D. py/o

QUICK QUIZ (Slide 8 of 8)

8. The combining form isch/o means:

- A. scanty
- B. to hold back; back
- C. night
- D. trigon